

Solow Growth Model ^{*}

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^{*}For more discussions, please refer to Barro and Sala-i-Martin [1], and Acemoglu [2].

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1 Introduction

Growth models have played important role for the modern development of macroeconomics. Starting from describing the dynamic behavior of Keynesian model (Harrod-Domar model), growth models are dynamic general equilibrium¹ models. Despite its simplicity, the Solow Growth Model is one of them. The difference between the Solow Growth Model and the Ramsey-Cass-Koopmans (RCK) Model is whether the saving rate is given, while the difference between the RCK Model and the Model of Endogenous Technological Progress is whether the technology is exogenously determined.

2 Social Planner's Economy

We will show that the allocations in a decentralized competitive market economy coincide with those in a social planner's economy.

2.1 The Economy

- The economy is an isolated island. There is a social planner.
- There is only one good, which can either be consumed after production or saved for investment at next period.
- Population grows at rate n , so $L(t) = e^{nt}L(0)$. Without loss of generality we assume that $L(0) = 1$.
- Each period, the social planner saves a constant fraction $s \in (0, 1)$ of contemporaneous output, which will be used as capital for the next period.
- Suppose that existing capital depreciates at a fixed rate $\delta \in [0, 1]$.

2.2 Production

- The production function for final good is written as

$$Y(t) = F(K(t), A(t)L(t)) \quad (1)$$

where $Y(t)$ is the total output of the final good, $K(t)$ is the capital stock, $L(t)$ is total employment, and $A(t)$ is technology at time t . We also assume that $F : R_+^3 \rightarrow R_+$ is twice differential in K and L .

- F satisfies "neoclassical" properties:

1. Constant returns to scale in K and L ,

$$F(\mu K(t), A(t)\mu L(t)) = \mu F(K(t), A(t)L(t))$$

¹Dynamic stochastic general equilibrium (DSGE) model is the main framework for modern macroeconomic analysis.

2. Positive and diminishing marginal products:

$$F_K(K, AL) \equiv \frac{\partial F(K, AL)}{\partial K} > 0, F_L(K, AL) \equiv \frac{\partial F(K, AL)}{\partial L} > 0$$

$$F_{KK}(K, AL) \equiv \frac{\partial^2 F(K, AL)}{\partial K^2} < 0, F_{LL}(K, AL) \equiv \frac{\partial^2 F(K, AL)}{\partial L^2} < 0$$

3. Inada conditions:

$$\lim_{K \rightarrow 0} F_K(K, AL) = \infty \text{ and } \lim_{K \rightarrow \infty} F_K(K, AL) = 0 \text{ for all } L > 0 \text{ and all } A.$$

$$\lim_{L \rightarrow 0} F_L(K, AL) = \infty \text{ and } \lim_{L \rightarrow \infty} F_L(K, AL) = 0 \text{ for all } K > 0 \text{ and all } A.$$

- Using Euler's Theorem², we can have the following result from the above assumptions:

$$Y(t) = F_L(K(t), A(t)L(t))L(t) + F_K(K(t), A(t)L(t))K(t)$$

- Production in intensive form:

$$\text{Define } k(t) \equiv \frac{K(t)}{A(t)L(t)} \text{ and } y(t) \equiv \frac{Y(t)}{A(t)L(t)}.$$

By CRS,

$$y = f(k) \equiv F(K, 1)$$

The properties of $f(k)$:

1. $f(0) = 0$
2. $f'(k) > 0 > f''(k)$
3. $\lim_{k \rightarrow 0} f(k) = \infty$ and $\lim_{k \rightarrow \infty} f(k) = 0$
4. $f'(k) \equiv \frac{df(k)}{dk} = \frac{dF(\frac{K}{AL}, 1)}{d\frac{K}{AL}} = \frac{\partial F(K, AL)}{\partial K}$
5. $\frac{\partial F(K, AL)}{\partial (AL)} = \frac{\partial (AL \times F(\frac{K}{AL}, 1))}{\partial (AL)} = f(k) - f'(k)k$

2.3 Law of Motions for Labor, Technology and Capital

- We have assumed $L(t) = e^{nt}L(0)$, $n \geq 0$, so $\dot{L}(t) \equiv \frac{dL(t)}{dt} = nL(t)$.
- By assuming $A(t) = e^{gt}A(0)$, $g \geq 0$, we get the growth rate of technology $\dot{A}(t) \equiv \frac{dA(t)}{dt} = gA(t)$.
- Optimally, $\dot{K}(t) = sY(t) - \delta K(t)$.

²Refer to the Mathematical Appendix M.B of Mas-Collel, Whinston and Green[3].

2.4 Dynamics of the Economy

How does the economy behave over time?

$$\dot{k}(t) = \frac{\dot{K}(t)}{A(t)L(t)} - \frac{K(t)}{[A(t)L(t)]^2} [A(t)\dot{L}(t) + L(t)\dot{A}(t)]$$

or

$$\dot{k}(t) = sf(k(t)) - (\delta + n + g)k(t) \quad (2)$$

2.5 Steady State

- In the steady state:

$$\dot{k}(t) = sf(k(t)) - (\delta + n + g)k(t) = 0$$

Thus

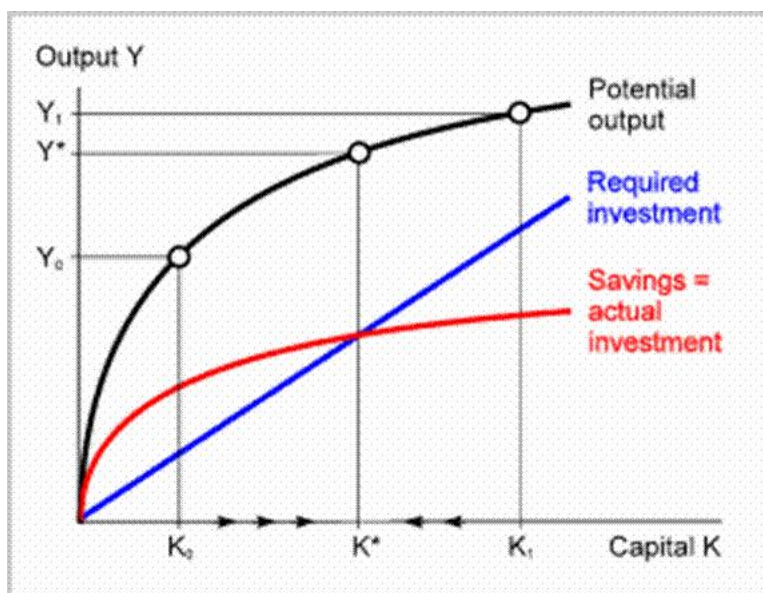
$$\frac{f(k^*)}{k^*} = \frac{(\delta + n + g)}{s}$$

- Balanced Growth Path:

1. Y and K grow at rate $n+g$;
2. $\frac{Y}{L}$ grows at rate g . Steady state growth rates in per capita income depend only on the rate of technological progress, g .

2.6 Transitional Dynamics

The growth rate is positive and decreases over time towards zero if $k_0 < k^*$; it is negative and increases over time towards zero if $k_0 > k^*$.



3 Market Economy

3.1 Households

- Households live an infinite amount of time. The number of heads in each household grow at constant rate $n \geq 0$.
- Each person in a household supplies one unit of labor inelastically.
- Households are assumed to save a constant exogenous fraction $s \in (0, 1)$ of their disposable income.

3.2 Firms

- Firms employ labor and rent capital in competitive labor and capital markets to produce the final good.
- The technology for each firm j is $Y_j(t) = F(K_j(t), A(t)L_j(t))$.
- The firms seek to maximize profits. The FOCs for an interior solution require

$$F_K(K_j(t), A(t)L_j(t)) = R_t$$

$$F_L(K_j(t), A(t)L_j(t)) = W_t$$

3.3 General Equilibrium

- Market Clearing:

$$\int_j K_j(t) dj = K(t) \text{ and } \int_j L_j(t) dj = L(t)$$

- Given the market structure, we know

$$R_j(t) = R_i(t), \text{ and } W_j(t) = W_i(t)$$

Thus all the firms have the same $\frac{K}{L}$ ratio. So the production function of the whole economy can be written as:

$$Y(t) = F(K(t), A(t)L(t)) \tag{3}$$

4 Further Topics

1. Consumption
2. Speed of adjustment
3. Government
4. Comparative Statics

References

- [1] Robert Barro, and Xavier Sala-i-Martin, 2004, *Economic Growth*, 2nd Edition, The MIT Press.
- [2] Darron Acemoglu, 2009, *Introduction to Modern Economic Growth*, Princeton University Press.
- [3] Andreu Mas-Collel, Michael Whinston, and Jerry R. Green, 1995, *Microeconomic Theory*, Oxford University Press.